

Exponential Algorithms for Knapsack

OPG-36311

L1 - TRACTABLE

PARTIALLY SOLVED

Computer Science

ORIGINAL PROBLEM STATEMENT

Conjecture

The famous 0-1 Knapsack problem is: Given $a_1, a_2, \dots, a_n$ and b integers, determine whether or not there are 0 – 1 values $x_1, x_2, \dots, x_n$ so that

$$\sum_{i=1}^n a_i x_i = b.$$

The best known worst-case algorithm runs in time $2^{n/2}$ times a polynomial in n . Is there an algorithm that runs in time $2^{n/3}$?